AMC 10-STYLE · FREE DIAGNOSTIC

AMC 10-Style Diagnostic Test

25	75 min	No
questions	time	calculator

Not affiliated with or endorsed by the Mathematical Association of America (MAA).

1. 25 multiple-choice problems, easy to hard. Each has one right answer: write its letter (A to E) in the box. No calculator.
2. 6 points for each right answer, 1.5 points for each blank and 0 for a wrong answer (150 in all). A guess pays only when you can rule out two or more choices.
3. Stuck? Skip it and come back. After the test, scan the code next to any problem you missed to watch the key move for it.

Name _____ Date _____ Score / 150

1 A museum charges $12 for an adult ticket and $8 for a student ticket. A group of 5 adults and 9 students pays with three $50 bills. How much change should the group get back?

(A) $12 **(B)** $18 **(C)** $22 **(D)** $28 **(E)** $32

Stuck? Scan for the key

Answer ☐

2 Three pears and two apples weigh 780 grams. Three pears and five apples weigh 1,140 grams. All the pears weigh the same, and all the apples weigh the same. How many grams does one pear weigh?

(A) 120 **(B)** 180 **(C)** 200 **(D)** 260 **(E)** 360

Stuck? Scan for the key

Answer ☐

3 Four identical printers print 1,200 pages in 15 minutes. At the same rate, how many minutes would 6 of these printers take to print 2,400 pages?

(A) 10 **(B)** 15 **(C)** 20 **(D)** 30 **(E)** 40

Stuck? Scan for the key

Answer ☐

4 At a market, 2 melons cost the same as 5 mangoes, and 3 mangoes cost the same as 4 lemons. One lemon costs 45 cents. How much do 3 melons cost?

(A) $3.00 **(B)** $3.60 **(C)** $4.05 **(D)** $4.50 **(E)** $5.40

Stuck? Scan for the key

Answer ☐

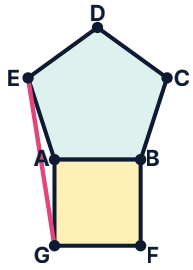
5 At a bakery, a muffin and a scone cost $5.10 together, a scone and a bagel cost $4.30 together, and a muffin and a bagel cost $4.60 together. How much does one scone cost?

(A) $1.90 **(B)** $2.00 **(C)** $2.15 **(D)** $2.40 **(E)** $2.70

Stuck? Scan for the key

Answer ☐

6 Regular pentagon ABCDE and square ABFG share the side AB, and the square lies outside the pentagon, as shown. What is the measure of angle AGE?



(A) 9° **(B)** 12° **(C)** 18° **(D)** 27° **(E)** 36°

Stuck? Scan for the key

Answer ☐

7 A list starts with 1. Each later number is the remainder when 3 times the number before it is divided by 11, so the list begins 1, 3, 9, 5, 4, ... What is the sum of the first 100 numbers in the list?

(A) 440 **(B)** 441 **(C)** 444 **(D)** 462 **(E)** 500

Stuck? Scan for the key

Answer ☐

8 The mean of 8 numbers is 15. Two of the numbers are removed, and the mean of the other 6 is 13. One removed number is 4 more than the other. What is the larger removed number?

(A) 19 **(B)** 20 **(C)** 21 **(D)** 22 **(E)** 23

Stuck? Scan for the key

Answer ☐

9 What is the sum of all odd positive multiples of 7 that are less than 300?

(A) 2940 **(B)** 3045 **(C)** 3085 **(D)** 3086 **(E)** 3087

Stuck? Scan for the key

Answer ☐

10 Eleven positive integers, not necessarily different, add up to 100. What is the greatest possible value of their median?

(A) 9 **(B)** 15 **(C)** 16 **(D)** 17 **(E)** 20

Stuck? Scan for the key

Answer ☐

11 A square with a whole-number side length has a smaller square, also with a whole-number side length, cut out of one corner. The L-shaped piece that is left has area 96. How many different side lengths could the original square have?

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3 **(E)** 4

Stuck? Scan for the key

Answer ☐

12 Lara multiplies 72 by a positive integer n. Her product turns out to be a perfect square and also a perfect cube. What is the smallest n she could have used?

(A) 648 **(B)** 649 **(C)** 650 **(D)** 651 **(E)** 1296

Stuck? Scan for the key

Answer ☐

13 A sequence begins 1, 4, 10, 19, 31, ... The differences between neighboring terms are 3, 6, 9, 12, and so on. What is the 20th term?

(A) 541 **(B)** 568 **(C)** 570 **(D)** 571 **(E)** 574

Stuck? Scan for the key

Answer ☐

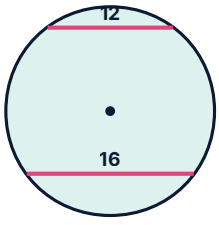
14 Each face of a cube is painted red or blue, each color equally likely and each face independently. What is the probability that no two red faces share an edge?

(A) 1/8 **(B)** 5/32 **(C)** 11/64 **(D)** 7/32 **(E)** 1/4

Stuck? Scan for the key

Answer ☐

15 In a circle of radius 10, two parallel chords have lengths 12 and 16. The center of the circle lies between the two chords. What is the distance between the chords?



(A) 2 **(B)** 10 **(C)** 12 **(D)** 14 **(E)** 16

Stuck? Scan for the key

Answer ☐

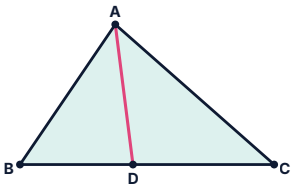
16 Call a three-digit number climbing if each digit is larger than the digit before it, like 268. How many climbing numbers have digits that add up to 15?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 12

Stuck?
Scan for the key

Answer ☐

17 In triangle ABC, AB = 12 and BC = 18. Point D lies on side BC so that angle BAD equals angle BCA. What is BD?

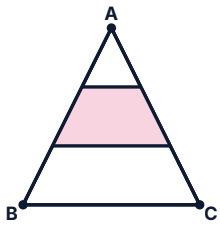


(A) 4 **(B)** 6 **(C)** 8 **(D)** 9 **(E)** 10

Stuck?
Scan for the key

Answer ☐

18 Triangle ABC has area 72. Two lines parallel to BC cut side AB into three equal parts and cut the triangle into three pieces. What is the area of the middle piece?



(A) 24 **(B)** 25 **(C)** 32 **(D)** 40 **(E)** 48

Stuck?
Scan for the key

Answer ☐

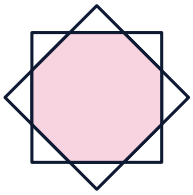
19 Which of the following numbers has exactly six positive divisors, with those six divisors adding up to 124?

(A) 44 **(B)** 45 **(C)** 75 **(D)** 76 **(E)** 98

Stuck?
Scan for the key

Answer ☐

20 Two squares with side length 2 have the same center. One is turned 45° from the other. What is the area of the region inside both squares?



(A) $2\sqrt{2}$ **(B)** 3 **(C)** $8\sqrt{2} - 8$ **(D)** $4\sqrt{2} - 2$ **(E)** $16 - 8\sqrt{2}$

Stuck?
Scan for the key

Answer ☐

21 For how many positive integers n is $\lfloor n/2 \rfloor + \lfloor n/3 \rfloor = n - 10$? (Here $\lfloor x \rfloor$ is the greatest integer that is not more than x.)

(A) 1 **(B)** 3 **(C)** 6 **(D)** 10 **(E)** 12

Stuck?
Scan for the key

Answer ☐

22 Mia rolls three fair dice and multiplies the three numbers. What is the probability that the result is divisible by 6?

(A) 83/216 **(B)** 1/2 **(C)** 125/216 **(D)** 7/12 **(E)** 133/216

Stuck?
Scan for the key

Answer ☐

23 Using all eight letters of COCONUTS, how many different strings can be made in which the two O's are separated by at least one other letter?

(A) 2520 **(B)** 7560 **(C)** 10080 **(D)** 15120 **(E)** 20160

Stuck?
Scan for the key

Answer ☐

24 How many integers n with $1 \le n \le 500$ make $n^3 - n$ a multiple of 120?

(A) 150 **(B)** 175 **(C)** 180 **(D)** 187 **(E)** 200

Stuck?
Scan for the key

Answer ☐

25 Square ABCD has side length 12. One circle passes through A and D and is tangent to side BC. A second circle passes through B and C and is tangent to side AD. The two circles meet at two points. What is the distance between those two points?

(A) $3\sqrt{6}$ **(B)** 12 **(C)** $6\sqrt{5}$ **(D)** $8\sqrt{3}$ **(E)** $6\sqrt{6}$

Stuck?
Scan for the key

Answer ☐

End of test. Check that every box has an answer.

AMC 10-STYLE · ANSWER KEY

Diagnostic Test: Answers

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Each problem uses one tool from the channel. For any you missed, scan the code next to that problem on the test to watch the lesson, then try the problem again tomorrow without looking. Score: 6 points for each right answer, 1.5 for each blank and 0 for each wrong answer (150 in all).

1 B $18 Tool #19: Circle the Ask

The ask is the change, not the cost. Cost: 5 × 12 + 9 × 8 = 132 dollars. Change: 150 − 132 = $18.

2 B 180

Tool #48: Take the Same Off Both Sides

Take the 3 pears and 2 apples off both: 3 apples weigh 1,140 − 780 = 360, so an apple is 120. Then 3 pears = 780 − 240 = 540: one pear is 180.

3 C 20 Tool #10: One Unit First

One printer: 1,200 ÷ 4 = 300 pages in 15 min, or 20 pages a minute. Six printers: 120 a minute. 2,400 ÷ 120 = 20 minutes.

4 D $4.50 Tool #2: Swap It

Swap down to lemons: 3 mangoes = 4 × 45 = 180¢, so a mango is 60¢. A melon is 5 × 60 ÷ 2 = 150¢. Three melons: $4.50.

5 D $2.40 Tool #40: Add All the Pairs

Add all three pairs: 5.10 + 4.30 + 4.60 = 14.00 counts each item twice, so one of each costs $7.00. Scone = 7.00 − 4.60 = $2.40.

6 A 9° Tool #60: Chase the Angles

At A: 360 − 108 − 90 = 162° between AE and AG. AE = AB = AG, so triangle AEG is isosceles: angle AGE = (180 − 162) ÷ 2 = 9°.

7 A 440 Tool #5: Find the Cycle

The list goes 1, 3, 9, 5, 4, then 1 again: it repeats every 5 numbers, and each block adds to 22. 100 numbers are 20 blocks: 20 × 22 = 440.

8 E 23 Tool #6: Averages Into Totals

Totals: 8 × 15 = 120 and 6 × 13 = 78, so the two removed add to 42. Two numbers 4 apart adding to 42: 19 and 23.

9 E 3087 Tool #16: First Plus Last

They are 7, 21, 35, ..., 287: 7 times the odd numbers 1 to 41, which is 21 terms. Sum = 21 × (7 + 287) ÷ 2 = 21 × 147 = 3,087.

10 B 15 Tool #4: Push to the Extreme

Push the five smallest down to 1 each (5 in all). The median and the five above it are each at least the median m, so 5 + 6m ≤ 100: m ≤ 15. Example: 1×5, 15×5, 20.

11 E 4 Tool #61: Factor the Difference

$m^2 - n^2 = (m - n)(m + n) = 96$, and both factors must be even: 2 × 48, 4 × 24, 6 × 16, 8 × 12. That gives m = 25, 14, 11 or 10: 4 sizes.

12 A 648 Tool #32: Prime Factor Split

$72 = 2^3 \times 3^2$. A square that is also a cube has every exponent a multiple of 6: $2^6 \times 3^6$. So n = $2^3 \times 3^4 = 648$.

13 D 571 Tool #34: Find the Rule

Term n = 1 + 3(1 + 2 + ... + (n − 1)) = 1 + 3 × (n − 1)n ÷ 2. For n = 20: 1 + 3 × 190 = 571.

14 B 5/32 Tool #59: Good Over Total

Two faces share an edge unless they are opposite. Good colorings: no red (1), one red (6), or one opposite pair red (3). 10 of 64 = 5/32.

15 D 14 Tool #57: Square the Sides

From the center to the 12-chord: $\sqrt{(10^2 - 6^2)} = 8$. To the 16-chord: $\sqrt{(10^2 - 8^2)} = 6$. Opposite sides of the center, so 8 + 6 = 14.

16 A 8 Tool #31: Organized List

List by the first digit: 159, 168; 249, 258, 267; 348, 357; 456. A first digit of 5 or more is too big. That is 8 numbers.

17 C 8 Tool #84: Find Similar Triangles

Triangles BAD and BCA share angle B and have angle BAD = angle BCA, so they are similar: BD/BA = BA/BC. BD = $12^2 \div 18 = 8$.

18 A 24 Tool #62: Scale Squared

The small triangles at A have sides 1/3 and 2/3 of ABC, so areas 72/9 = 8 and 72 × 4/9 = 32. Middle piece: 32 − 8 = 24.

19 C 75 Tool #45: Try the Choices

Six divisors means p^2q or p^5. Check: 45 sums to 78, 50 to 93, 63 to 104. $75 = 5^2 \times 3$: (1 + 5 + 25)(1 + 3) = 124.

20 C $8\sqrt{2} - 8$ Tool #44: Cut It Into Pieces

The overlap is a regular octagon: the square minus 4 corner triangles. Each corner has legs $2 - \sqrt{2}$, so the area is $4 - 2(2 - \sqrt{2})^2 = 8\sqrt{2} - 8$.

21 C 6 Tool #82: Split Into Cases

Split by the remainder r of n mod 6: n = 6k + r. The left side is 5k plus 0, 0, 1, 2, 3, 3 for r = 0 to 5. Each r gives one k: n = 60, 55, 56, 57, 58, 53.

22 E 133/216 Tool #1: Whole Minus Part

Whole minus the bad rolls. No even number: 1/8. No multiple of 3: 8/27. Both happen (all 1s and 5s): 1/27, counted twice. 1 − 1/8 − 8/27 + 1/27 = 133/216.

23 B 7560 Tool #58: Line Up, Then Divide

All strings: 8! ÷ (2! × 2!) = 10,080 (divide for the two C's and two O's). Glue the O's: 7! ÷ 2! = 2,520. 10,080 − 2,520 = 7,560.

24 D 187 Tool #85: Work in Mod

$n^3 - n = (n - 1)n(n + 1)$ is a multiple of 3. Mod 5, n must be 0, 1 or 4. Mod 8, n is odd or a multiple of 8. That is 15 of every 40; 12 × 15 + 7 = 187.

25 E $6\sqrt{6}$ Tool #83: Set Up Coordinates

Put A at (0, 0) and D at (0, 12). The first center is (h, 6) with $h^2 + 36 = (12 - h)^2$, so h = 4.5 and r = 7.5; the second is (7.5, 6). Chord: $2\sqrt{(7.5^2 - 1.5^2)} = 6\sqrt{6}$.